

ELLSWORTH AFB, SD, USA

WMO: 726625

Lat: 44.150N

Lon: 103.100W

Elev: 999

StdP: 89.88

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24006

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.0	-19.0	-24.8	0.4	-21.1	-22.2	0.6	-18.5	18.7	-1.6	17.2	-3.9	4.0	10	0.657

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	35.2	19.2	33.0	18.9	31.2	18.7	22.2	29.3	21.2	28.5	20.3	27.7	5.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.3	16.9	24.8	19.0	15.6	24.4	17.8	14.4	23.6	70.2	29.0	66.3	28.4	62.7	27.8	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.8	13.5	11.6	-25.3	38.4	3.2	2.1	-27.6	39.9	-29.5	41.2	-31.3	42.3	-33.6	43.9		
			-25.6	24.2	3.1	2.1	-27.8	25.7	-29.6	26.9	-31.3	28.0	-33.6	29.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	-3.8	-2.7	2.1	7.0	12.6	18.8	22.9	22.4	17.4	9.0	2.2	-3.6
	DBStd	11.28	7.35	7.79	7.45	5.58	4.73	4.25	3.37	3.43	5.09	5.50	6.74	7.26
	HDD10.0	1943	429	357	255	121	27	2	0	0	7	83	242	421
	HDD18.3	3890	687	588	502	342	186	43	4	7	75	291	484	679
	CDD10.0	1491	1	2	11	29	107	266	399	386	228	52	9	1
	CDD18.3	396	0	0	0	1	8	58	146	134	47	2	0	0
	CDH23.3	4681	0	0	9	35	125	692	1609	1487	655	64	4	0
	CDH26.7	2016	0	0	1	6	32	269	733	690	266	17	1	0
Wind	WSAvg	4.8	5.0	5.2	5.3	5.6	5.2	4.6	4.1	4.1	4.4	5.0	4.7	4.7
Precipitation	PrecAvg	450	9	10	23	55	89	73	55	47	32	37	12	8
	PrecMax	601	26	23	48	122	186	161	99	107	76	136	33	27
	PrecMin	310	0	0	1	10	38	17	15	20	5	2	1	0
	PrecStd	88	7	7	13	26	44	40	24	22	18	32	9	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.7	19.7	24.2	27.5	30.2	36.8	37.5	37.5	36.3	29.2	22.9	16.6
		MCWB	6.1	8.5	10.6	12.9	17.2	18.7	19.8	19.3	18.2	14.3	10.0	6.8
	2%	DB	11.9	14.8	20.1	23.2	26.7	32.3	34.9	34.9	32.3	24.9	18.2	12.2
		MCWB	5.1	6.2	9.2	11.8	15.7	18.9	20.1	18.8	17.2	12.6	8.1	5.0
	5%	DB	9.0	11.3	16.6	19.8	23.9	29.9	32.8	32.7	29.9	21.4	15.1	8.7
		MCWB	3.4	4.6	7.6	10.6	14.5	18.6	19.7	18.6	16.7	11.2	6.8	3.1
	10%	DB	6.5	7.8	13.1	16.8	21.5	27.4	30.9	30.8	27.2	18.0	12.2	5.9
		MCWB	2.0	2.7	6.1	9.0	13.7	17.8	19.5	18.6	15.5	10.0	5.5	1.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.1	8.8	11.7	14.3	19.5	23.1	24.1	23.3	20.9	15.6	10.6	7.6
		MCDB	14.3	18.9	22.0	23.6	26.3	29.1	29.6	29.4	28.9	25.3	22.2	16.1
	2%	WB	5.1	6.7	9.6	12.4	17.2	21.1	22.5	21.6	19.1	13.7	8.6	5.2
		MCDB	11.0	14.2	19.4	21.5	24.2	27.9	29.9	28.8	27.7	22.1	17.3	11.5
	5%	WB	3.7	4.9	8.0	11.0	15.6	19.9	21.7	20.5	17.9	12.0	7.2	3.3
		MCDB	8.5	10.6	15.9	18.6	22.2	27.3	29.0	28.4	27.0	19.6	14.6	8.1
	10%	WB	2.3	3.0	6.4	9.6	14.3	18.7	20.7	19.6	16.6	10.5	5.6	1.7
		MCDB	6.4	7.4	12.7	16.0	19.7	26.1	28.0	27.5	25.3	17.5	11.7	5.6
Mean Daily Temperature Range	5% DB	MDBR	11.5	11.7	12.6	12.3	11.9	12.9	13.6	14.3	14.5	12.6	12.2	11.0
		MCDBR	15.2	16.6	18.4	18.4	16.2	16.9	16.5	17.5	18.6	18.1	17.1	15.0
		MCWBR	9.7	9.6	9.8	9.5	8.3	7.4	6.1	6.6	7.2	8.7	9.3	9.2
	5% WB	MCDBR	14.5	15.5	17.2	16.5	14.5	15.0	13.8	14.8	16.0	16.0	16.3	14.2
		MCWBR	9.5	9.3	9.5	8.9	7.8	7.8	6.6	6.7	6.7	8.1	9.2	9.1
Clear-Sky Solar Irradiance	taub	0.252	0.261	0.279	0.313	0.331	0.344	0.366	0.361	0.325	0.288	0.262	0.251	
	taud	2.579	2.552	2.552	2.441	2.467	2.456	2.425	2.419	2.504	2.578	2.565	2.564	
	Ebn at Noon	907	953	971	952	939	924	898	891	904	905	883	873	
	Edh at Noon	62	76	88	107	108	109	112	108	92	74	63	57	
All-Sky Solar Radiation	RadAvg	1.91	2.88	4.04	5.02	5.96	6.91	7.04	6.05	4.78	3.14	2.12	1.60	
	RadStd	0.10	0.17	0.31	0.30	0.58	0.52	0.26	0.27	0.30	0.30	0.16	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page